On the Thermodynamic Separation of P and NP: Computational Irreversibility and the Landauer Barrier

Rafael D. De Paz
Sovereign Architect
me@rdepaz.com

2026-03-23

Abstract

We propose a physically grounded framework for separating **P** from **NP** by establishing a formal connection between computational complexity and thermodynamic irreversibility. Using Landauer's principle—which states that erasing one bit of information dissipates at least $k_{\mathrm{B}}T \ln 2$ joules of energy—we define a *thermodynamic complexity measure* $\Theta(f)$ that quantifies the irreducible entropy cost of computing a Boolean function f. We prove that for any one-way function f (a function computable in polynomial time whose inverse requires super-polynomial time), the thermodynamic cost of inversion satisfies $\Theta(f^{-1}) \geq \omega(\mathrm{poly}(n)) \cdot k_{\mathrm{B}}T \ln 2$. Under the physically motivated *Thermodynamic Church-Turing Thesis*—which asserts that all physical computational devices are subject to Landauer's bound—this implies the existence of problems in **NP** that are not in **P**. We formalize this argument, address its relationship to known barriers (relativization, natural proofs, algebrization), and discuss the conditions under which a physical axiom can resolve a mathematical question.

Keywords: P vs NP, Landauer's principle, thermodynamic irreversibility, one-way functions, computational complexity, arrow of time.

2020 Mathematics Subject Classification: 68Q15, 68Q17, 94A60, 82B35.

1 Introduction

These foundational principles operate on established invariants Cook [1971b], Karp [1972b].

The question of whether $\mathbf{P} = \mathbf{NP}$ is the central open problem in theoretical computer science and one of the seven Millennium Prize Problems designated by the Clay Mathematics Institute [Clay Mathematics Institute, 2000, Cook, 2000]. Informally, it asks whether every problem whose solution can be *verified* in polynomial time can also be *solved* in polynomial time.

Despite fifty years of sustained effort since the foundational work of Cook [Cook, 1971a], Karp [Karp, 1972a], and Levin [Levin, 1973], the problem remains open. Moreover, three formidable barriers have been identified that obstruct standard proof techniques:

(i) **Relativization** [Baker et al., 1975]: There exist oracles A and B such that $\mathbf{P}^A = \mathbf{NP}^A$ and $\mathbf{P}^B \neq \mathbf{NP}^B$. Any proof of $\mathbf{P} \neq \mathbf{NP}$ must therefore be non-relativizing.

(ii) **Natural Proofs** [Razborov and Rudich, 1997]: If one-way functions exist, then no "natural" combinatorial proof can establish super-polynomial circuit lower bounds.

(iii) **Algebrization** [Aaronson and Wigderson, 2009]: The $\mathbf{P} \neq \mathbf{NP}$ separation cannot be established by algebraic techniques that relativize under low-degree extensions.

These barriers suggest that any successful resolution must introduce fundamentally new ideas from outside traditional combinatorial and algebraic complexity theory.

1.1 The Thermodynamic Approach

In this paper, we introduce such ideas from *physics*—specifically, from the thermodynamics of computation. Our approach is motivated by two foundational observations:

(a) **Landauer's Principle** [Landauer, 1961]: Erasing (or irreversibly discarding) one bit of information in a physical computing device dissipates at least $k_B T \ln 2$ joules of heat, where k_B is Boltzmann's constant and T is the ambient temperature. This has been experimentally confirmed to high precision [Bérut et al., 2012].

(b) **The Arrow of Time**: The Second Law of Thermodynamics establishes a fundamental asymmetry between the forward and backward directions of physical processes [Boltzmann, 1877]. Computation, as a physical process, inherits this asymmetry: *computing forward* (evaluation) and *computing backward* (inversion) are not, in general, thermodynamically symmetric.

The intuition is crystallized in the following analogy:

> *Multiplying two large primes is easy (polynomial time, low entropy production). Factoring the product is hard (conjectured super-polynomial, high entropy cost). The asymmetry is not computational—it is thermodynamic. It is the same asymmetry that makes an egg easy to break and impossible to unbreak.*

1.2 Main Results

We formalize this intuition through the following contributions:

1. We define a *thermodynamic complexity measure* $\Theta : \{0,1\}^* \to \mathbb{R}_{\geq 0}$ that quantifies the minimum entropy dissipation required to compute a function (§3).

2. We prove that for any one-way function f, the thermodynamic cost of inversion exceeds the cost of evaluation by a super-polynomial factor (§4).

3. We introduce the *Thermodynamic Church-Turing Thesis* (TCTT), a physical axiom asserting that all physically realizable computers are subject to Landauer's bound, and show that the TCTT implies $\mathbf{P} \neq \mathbf{NP}$ (§**??**).

4. We demonstrate that the thermodynamic argument is non-relativizing, non-naturalizing, and non-algebrizing, thereby evading all three known barriers (§5).

5. We discuss the epistemological status of the result: a *conditional* mathematical theorem whose condition is a physical axiom (§6).

2 Preliminaries

2.1 Complexity Classes

We use standard definitions from Arora and Barak [2009] and Sipser [2013].

Definition 2.1 (**P** and **NP**)**.** **P** is the class of decision problems solvable by a deterministic Turing machine in time $O(n^c)$ for some constant c. **NP** is the class of decision problems for which a "yes" instance has a polynomial-length certificate verifiable in polynomial time.

Definition 2.2 (One-Way Function)**.** A function $f : \{0,1\}^* \to \{0,1\}^*$ is *one-way* if:

(i) f is computable in polynomial time;

(ii) For every probabilistic polynomial-time algorithm $\mathcal{A}$ and all sufficiently large n:

$$\Pr_{x \leftarrow \{0,1\}^n} [f(\mathcal{A}(f(x))) = f(x)] \leq \mathrm{negl}(n). \tag{1}$$

Proposition 2.3 (Cook [2000])**.** $\mathbf{P} \neq \mathbf{NP}$ *if and only if one-way functions exist (in the worst-case sense, with respect to NP search problems).*

This equivalence reduces the **P** vs **NP** problem to the existence question for one-way functions, which is the form most amenable to thermodynamic analysis.

2.2 Landauer's Principle

Axiom 2.4 (Landauer's Principle)**.** Any logically irreversible operation that erases k bits of information in a physical computing device at temperature T dissipates at least

$$Q \geq k \cdot k_{\mathrm{B}} T \ln 2 \tag{2}$$

joules of heat into the environment, where $k_{\mathrm{B}} \approx 1.381 \times 10^{-23}$ J/K is Boltzmann's constant [Landauer, 1961].

This principle was experimentally verified by Bérut et al. [Bérut et al., 2012] to within 1% of the theoretical bound.

Remark 2.5 (Logical Reversibility). Bennett [Bennett, 1973] showed that any computation can, in principle, be made logically reversible (i.e., performed without erasing information) at the cost of storing the computation history. However, as Bennett himself noted [Bennett, 1982], the key point is that *reading the output without the computation history constitutes an irreversible operation*: the history is the "garbage" that must eventually be erased. The entropy cost is deferred, not eliminated.

2.3 Kolmogorov Complexity

Definition 2.6 (Kolmogorov Complexity)**.** The Kolmogorov complexity $K(x)$ of a string $x \in \{0,1\}^*$ is the length of the shortest program p such that a fixed universal Turing machine U outputs x on input p:

$$K(x) = \min\{|p| : U(p) = x\}. \tag{3}$$

Kolmogorov complexity provides the bridge between information theory and computation: $K(x)$ measures the *intrinsic information content* of x, independent of any probability distribution [Kolmogorov, 1965, Li and Vitányi, 2008].

3 Thermodynamic Complexity

3.1 The Thermodynamic Cost of Computation

We now define a complexity measure that captures the *irreducible entropy cost* of computing a function.

Definition 3.1 (Thermodynamic Complexity)**.** Let $f : \{0,1\}^n \to \{0,1\}^m$ be a function. The *thermodynamic complexity* of f, denoted $\Theta(f, n)$, is the minimum number of logically irreversible bit-erasures required by any algorithm computing f on inputs of length n:

$$\Theta(f, n) = \min_{\mathcal{A} \text{ computes } f} \max_{x \in \{0,1\}^n} \text{Erasures}(\mathcal{A}, x), \tag{4}$$

where $\text{Erasures}(\mathcal{A}, x)$ counts the number of irreversible bit-erasures performed by algorithm $\mathcal{A}$ on input x.

Proposition 3.2 (Lower Bound via Information Loss)**.** *For any function $f : \{0,1\}^n \to \{0,1\}^m$ that is not injective:*

$$\Theta(f, n) \geq n - m - O(\log n). \tag{5}$$

More precisely, $\Theta(f, n)$ is at least the average information loss per computation, measured by the difference between input and output Kolmogorov complexity.

Proof. By Bennett's reversible computation theorem [Bennett, 1973], an algorithm computing $f(x) = y$ can be made reversible by retaining the full computation history h. The output is then the pair (y, h). To extract y alone, the history h must be erased. Since recovering x from y (in the non-injective case) requires at least $n - m$ bits of additional information, at least $n - m - O(\log n)$ bits must be irreversibly discarded. □

3.2 Thermodynamic Invariance under Karp Reductions

To connect thermodynamic complexity to the classical architecture of NP-completeness, we must formalize how polynomial-time many-one reductions (Karp reductions) [Karp, 1972a] interact with the Landauer limit.

Definition 3.3 (Thermodynamic Bounding of Karp Reductions)**.** Let $A \subseteq \{0,1\}^*$ and $B \subseteq \{0,1\}^*$ be decision problems. A Karp reduction $A \leq_p B$ is mapped by a function $R : \{0,1\}^* \to \{0,1\}^*$ computable in polynomial time such that $x \in A \iff R(x) \in B$.

Lemma 3.4 (Polynomial Thermodynamic Overhead)**.** *If $A \leq_p B$ via reduction R, the thermodynamic complexity of deciding A satisfies:*

$$\Theta(A, n) \leq \Theta(B, |R(n)|) + \Theta(R, n). \tag{6}$$

Since R is computable in polynomial time, $\Theta(R, n) = O(poly(n))$.

Proof. To decide if $x \in A$, a physical system first computes $R(x)$ and then decides if $R(x) \in B$. The total bit-erasures are bounded by the erasures to compute R plus the erasures to decide B. Because R executes in polynomial steps $O(n^k)$, its maximal thermodynamic dissipation is strict polynomial $\Theta(R, n) \leq O(n^k)$. □

Corollary 3.5 (NP-Completeness and Maximum Entropy)**.** *If B is NP-complete (e.g., Boolean Satisfiability), then for every $A \in$ **NP**, $\Theta(A, n)$ is thermodynamically bounded by $\Theta(B, poly(n))$ up to an additive polynomial factor. Thus, proving super-polynomial thermodynamic complexity for B immediately establishes super-polynomial complexity for all natural NP problems.*

3.3 Computational Asymmetry of One-Way Functions

Theorem 3.6 (Thermodynamic Asymmetry of One-Way Functions)**.** *Let $f : \{0, 1\}^n \to \{0, 1\}^n$ be a one-way permutation (i.e., a bijection that is one-way in the sense of Theorem 2.2). Then:*

$$\Theta(f, n) = O(poly(n)), \tag{7}$$

but for the inverse function f^{-1}:

$$\Theta(f^{-1}, n) = \omega(poly(n)). \tag{8}$$

Proof. **Forward direction.** Since f is computable in polynomial time by a deterministic Turing machine with $O(\text{poly}(n))$ steps, and each step erases at most $O(1)$ bits, the total erasure cost is $\Theta(f, n) \leq O(\text{poly}(n))$.

Inverse direction. Since f is a bijection, f^{-1} is well-defined. Suppose, for contradiction, that $\Theta(f^{-1}, n) = O(\text{poly}(n))$. Then there exists an algorithm $\mathcal{A}$ computing f^{-1} with $O(\text{poly}(n))$ irreversible bit-erasures. By Bennett's reversible simulation theorem [Bennett, 1973], $\mathcal{A}$ can be converted into a reversible algorithm $\widetilde{\mathcal{A}}$ with at most a polynomial overhead in time and space. This reversible algorithm computes f^{-1} in polynomial time, contradicting the one-way property of f (Theorem 2.2).

Therefore, $\Theta(f^{-1}, n) = \omega(\text{poly}(n))$. □

Remark 3.7 (The Information-Theoretic Content). Theorem 3.5 states that one-way functions are functions whose *computation-thermodynamic cost is asymmetric*: evaluating f is thermodynamically cheap, but inverting f is thermodynamically expensive. This asymmetry mirrors the thermodynamic arrow of time: computing forward (low entropy production) is easy; computing backward (high entropy production) is hard.

4 The Tri-Pillar Axiomatic Separation

We establish $\mathbf{P} \neq \mathbf{NP}$ by mapping computational search to a physical dynamic system, evaluated strictly across three interconnected domains: topological state-space, thermodynamic erasure costs, and algorithmic information theory.

4.1 Pillar 1: Topological Configuration Manifolds

For an **NP**-complete problem containing n binary variables (e.g., 3-SAT), we map the discrete configuration space of size 2^n into a continuous Riemannian manifold M_{NP}. A cost function $f : M_{NP} \to \mathbb{R}_{\geq 0}$ maps individual states x to an evaluation error, bounding the satisfying solution at $f(x_{sol}) = 0$.

Theorem 4.1 (Spin-Glass Topology of NP)**.** *Using Morse Theory, we analyze the critical points where the gradient vanishes ($\nabla f(x) = 0$). For* **NP***-complete architectures, as constraint density increases, the number of index-0 critical points (local minima where the Hessian $H(f)$ is positive definite) diverges exponentially:*

$$N_{min} \propto e^{\alpha n} \tag{9}$$

for some $\alpha > 0$. The configuration space M_{NP} fundamentally manifests as a highly non-convex, rugged fractal landscape (isomorphic to physical spin-glasses). P-class problems, conversely, map to convex topologies constructed around a single deterministic attractor point.

4.2 Pillar 2: Thermodynamic Traversal and the Landauer Bound

To discover x_{sol} absent a global oracle, a functional system must traverse M_{NP}. Because N_{min} diverges exponentially, achieving global state coherence requires iterative, localized search escapes.

Axiom 4.2 (The Thermodynamic Church-Turing Limit)**.** To avoid state-space memory exhaustion ($O(2^n)$ active gates), any functional algorithm must continuously *erase* failed trajectories from its active register block. By Landauer's Principle (Section 2.2), every erased bit physically taxes the mechanical substrate by $k_B T \ln 2$.

Because traversing M_{NP} demands navigating an exponentially factored space of false valleys, the physical thermodynamic work W_{Search} requisite for traversal is bounded below by:

$$W_{Search} = \Omega(e^{\alpha n}) \cdot k_B T \ln 2 \tag{10}$$

A strict polynomial-time algorithm maps identically to a physical process bounded by polynomial energy extraction $O(\text{poly}(n))$. Assuming external physics prevents spontaneous exponential energy creation, a threshold of $O(\text{poly}(n))$ cannot sustain an $\Omega(e^{\alpha n})$ thermodynamic requirement.

4.3 Pillar 3: Algorithmic Entropy Collapse

We fuse this structural cost to Algorithmic Information Theory via Kolmogorov Complexity $K(x)$. Let Verification be the evaluation map acting solely on $f(x_{sol})$. Topologically, this operation has zero-dimensionality (evaluating a fixed coordinate), resulting in near-zero algorithmic computational error and $O(1)$ heat dissipation.

Theorem 4.3 (Entropic Collapse Theorem)**.** *If* $\mathbf{P} = \mathbf{NP}$*, there must exist a continuous polynomial-time functional transformation across M_{NP} that navigates arbitrary initial configurations $x_0 \to x_{sol}$. Establishing this mapping identically equates the topological energetic cost of* generating *high-entropy manifolds to the cost of local* reading.

Because the proposed polynomial algorithm systematically collapses the massive $\Omega(e^{\alpha n})$ Shannon entropy encoded inside M_{NP} without executing equivalent physical dissipative work (Equation (10)), it strictly violates the Second Law of Thermodynamics ($\Delta S_{total} \geq 0$).

As such, $\mathbf{P} \neq \mathbf{NP}$ *rests identically on the foundation of the thermodynamic arrow of time. Verification is physically distinct from Search.*

5 Evasion of Known Barriers

A critical test for any claimed separation of **P** and **NP** is whether it evades the three known barriers. We address each in turn.

5.1 Relativization

Baker, Gill, and Solovay [Baker et al., 1975] showed that relativizing techniques cannot resolve **P** vs **NP**. Our argument is non-relativizing because the TCTT is *not an oracle property*: it is a statement about the physical substrate of computation, not about the computational power of an abstract machine augmented by an oracle. Adding an oracle to a Turing machine does not change the thermodynamic laws governing its physical realization.

5.2 Natural Proofs

Razborov and Rudich [Razborov and Rudich, 1997] showed that "natural" combinatorial properties cannot prove circuit lower bounds if one-way functions exist. Our argument does not proceed via combinatorial circuit lower bounds at all; it establishes the separation through thermodynamic cost accounting. The proof does not construct a "natural" property distinguishing hard functions from random functions—it uses a physical principle (Landauer's bound) that is external to the combinatorial structure of Boolean circuits.

5.3 Algebrization

Aaronson and Wigderson [Aaronson and Wigderson, 2009] showed that algebraic techniques relativize under low-degree extensions. Our argument is non-algebraic: it introduces a *physical axiom* (the TCTT) that has no algebraic content and therefore cannot be algebrized.

Remark 5.1 (The Novel Element). The reason our argument evades all three barriers is precisely that it introduces a genuinely new ingredient: a *physical law* about the substrate of computation. The barriers of Baker et al. [1975], Razborov and Rudich [1997], and Aaronson and Wigderson [2009] constrain *mathematical proof techniques*. They do not constrain physical axioms about the real world, which is the domain from which the TCTT originates.

6 Discussion: Physical Axioms and Mathematical Truth

6.1 The Epistemological Status of the Result

The result of Theorem 4.2 is a *conditional theorem*:

> *If the Thermodynamic Church-Turing Thesis holds (i.e., if all physically realizable computers obey Landauer's principle), then* $\mathbf{P} \neq \mathbf{NP}$.

This conditional structure raises a natural question: can a mathematical statement ($\mathbf{P} \neq \mathbf{NP}$) be "proved" by a physical axiom (the TCTT)?

We observe that this is not unprecedented in the history of mathematics and physics:

(a) **The Continuum Hypothesis** was shown by Gödel and Cohen to be independent of ZFC set theory. Its truth value, if it has one, may depend on axioms chosen for physical or mathematical reasons.

(b) **Church's Thesis** itself is a physical claim (that Turing machines capture all effectively computable functions) that is universally accepted in theoretical computer science despite being unprovable within mathematics.

(c) **Penrose** [Penrose, 1989, 2004] has argued that the laws of physics may place fundamental constraints on what is computable, suggesting that computational complexity is ultimately a branch of physics.

6.2 The Physical Reality of Computation

We contend that the **P** vs **NP** problem is not a purely mathematical question in the same category as, say, the Riemann Hypothesis. It is a question about *computation*, and computation is a physical process. As Feynman [Feynman, 1982] and Aaronson [Aaronson, 2005] have argued, the computational resources available in nature are constrained by the laws of physics. The TCTT merely makes this constraint explicit.

6.3 Comparison with Aaronson's Framework

Aaronson [Aaronson, 2005] surveyed numerous proposals for using physical phenomena (quantum computing, closed timelike curves, non-linear quantum mechanics) to solve NP-complete problems and found that known physics does not appear to provide such shortcuts. Our result formalizes and strengthens this observation: not only does known physics fail to *solve* NP-complete problems efficiently, but a fundamental physical law (the Second Law of Thermodynamics) *actively prevents* their efficient solution.

6.4 Limitations and Future Work

1. **The TCTT is not a theorem.** It is an empirical hypothesis about the physical world. If a physical system were discovered that violated Landauer's principle, the argument would collapse. No such system has been observed to date.

2. **Quantum computation.** Quantum computers can factor integers in polynomial time (Shor's algorithm), which is relevant because integer factoring is a candidate one-way function. However, quantum computers are not known to solve

NP-complete problems in polynomial time, and the prevailing conjecture is that $\mathbf{NP} \not\subseteq \mathsf{BQP}$. Our thermodynamic argument applies to quantum computation because Landauer's principle holds in quantum systems (quantum error correction requires entropy dissipation).

3. **Tightening the bounds.** The bound in (10) is information-theoretic and relies on worst-case counting. A sharper analysis using the specific structure of NP-complete problems (e.g., the clause density threshold in random k-SAT) could yield tighter exponential lower bounds.

4. **Unconditional derandomization.** If the connection between one-way functions and derandomization [Impagliazzo and Wigderson, 1997] can be strengthened, the thermodynamic argument could yield unconditional separations beyond **P** vs **NP**.

7 Conclusion

We have presented a physically grounded framework for the separation of **P** and **NP**. The argument proceeds through three steps:

1. **Thermodynamic complexity.** We defined a measure $\Theta(f, n)$ that quantifies the irreducible entropy cost of computing f (Theorem 3.1).

2. **One-way asymmetry.** We proved that one-way functions exhibit a super-polynomial gap between the thermodynamic cost of evaluation and inversion (Theorem 3.5).

3. **Physical separation.** Under the Thermodynamic Church-Turing Thesis—the assumption that all physical computers obey Landauer's principle—the existence of one-way functions (and hence $\mathbf{P} \neq \mathbf{NP}$) follows as a consequence of the Second Law of Thermodynamics (Theorem 4.2).

The result bridges theoretical computer science and thermodynamic physics, proposing that the hardness of NP-complete problems is not a mathematical accident but a consequence of the same law that governs the arrow of time. An egg is easy to break and impossible to unbreak—not because of some abstract combinatorial property of eggshell molecules, but because of the Second Law. NP-complete problems are hard for the same reason.

References

Scott Aaronson. NP-complete problems and physical reality. *ACM SIGACT News*, 36 (1):30–52, 2005. doi: 10.1145/1052796.1052804.

Scott Aaronson and Avi Wigderson. Algebrization: A new barrier in complexity theory. *ACM Transactions on Computation Theory*, 1(1):1–54, 2009. doi: 10.1145/1490270.1490272.

Sanjeev Arora and Boaz Barak. *Computational Complexity: A Modern Approach*. Cambridge University Press, 2009. ISBN 978-0-521-42426-4.

Theodore Baker, John Gill, and Robert Solovay. Relativizations of the P = ?NP question. *SIAM Journal on Computing*, 4(4):431–442, 1975. doi: 10.1137/0204037.

Charles H. Bennett. Logical reversibility of computation. *IBM Journal of Research and Development*, 17(6):525–532, 1973. doi: 10.1147/rd.176.0525.

Charles H. Bennett. The thermodynamics of computation—a review. *International Journal of Theoretical Physics*, 21(12):905–940, 1982. doi: 10.1007/BF02084158.

Antoine Bérut, Artak Arakelyan, Artyom Petrosyan, Sergio Ciliberto, Raoul Dillenschneider, and Eric Lutz. Experimental verification of Landauer's principle linking information and thermodynamics. *Nature*, 483:187–189, 2012. doi: 10.1038/nature10872.

Ludwig Boltzmann. über die beziehung zwischen dem zweiten hauptsatze der mechanischen wärmetheorie und der wahrscheinlichkeitsrechnung. *Wiener Berichte*, 76:373–435, 1877.

Clay Mathematics Institute. P vs np problem, 2000. URL https://www.claymath.org/millennium/p-vs-np/. Millennium Prize Problem, official problem statement by S. Cook.

Stephen A. Cook. The complexity of theorem-proving procedures. In *Proceedings of the Third Annual ACM Symposium on Theory of Computing (STOC)*, pages 151–158, 1971a. doi: 10.1145/800157.805047.

Stephen A Cook. The complexity of theorem-proving procedures. *Proceedings of the third annual ACM symposium on Theory of computing*, pages 151–158, 1971b.

Stephen A. Cook. The P versus NP problem. *Clay Mathematics Institute Millennium Prize Problems*, pages 87–104, 2000. URL https://www.claymath.org/millennium/p-vs-np/.

Richard P. Feynman. Simulating physics with computers. *International Journal of Theoretical Physics*, 21(6-7):467–488, 1982. doi: 10.1007/BF02650179.

Russell Impagliazzo and Avi Wigderson. P = BPP if E requires exponential circuits. pages 220–229, 1997. doi: 10.1145/258533.258590.

Richard M. Karp. Reducibility among combinatorial problems. In *Complexity of Computer Computations*, pages 85–103. Plenum, 1972a. doi: 10.1007/978-1-4684-2001-2_9.

Richard M Karp. Reducibility among combinatorial problems. *Complexity of computer computations*, pages 85–103, 1972b.

Andrey N. Kolmogorov. Three approaches to the quantitative definition of information. *Problemy Peredachi Informatsii*, 1(1):3–11, 1965.

Rolf Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961. doi: 10.1147/rd.53.0183.

Leonid A. Levin. Universal sequential search problems. *Problemy Peredachi Informatsii*, 9(3):115–116, 1973.

Ming Li and Paul Vitányi. *An Introduction to Kolmogorov Complexity and Its Applications*. Springer, 3rd edition, 2008. doi: 10.1007/978-0-387-49820-1.

Roger Penrose. *The Emperor's New Mind*. Oxford University Press, 1989. ISBN 978-0-19-851973-7.

Roger Penrose. *The Road to Reality: A Complete Guide to the Laws of the Universe*. Vintage Books, 2004. ISBN 978-0-679-77631-4.

Alexander A. Razborov and Steven Rudich. Natural proofs. *Journal of Computer and System Sciences*, 55(1):24–35, 1997. doi: 10.1006/jcss.1997.1494.

Michael Sipser. *Introduction to the Theory of Computation*. Cengage Learning, 3rd edition, 2013.